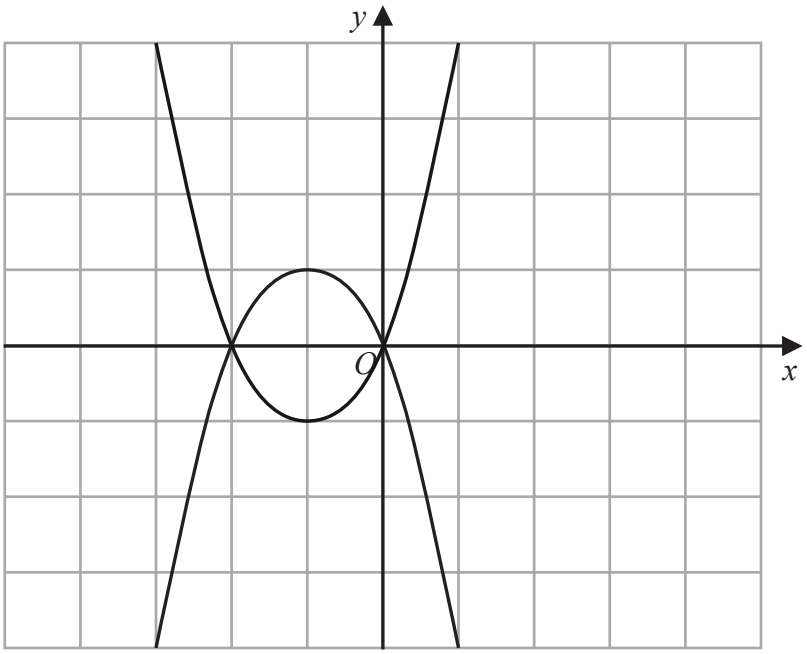
21 The graph of $y = f(x)$ is shown on the grid below.



1 mark
Reflect in the x-axis.
Same x; opposite y.

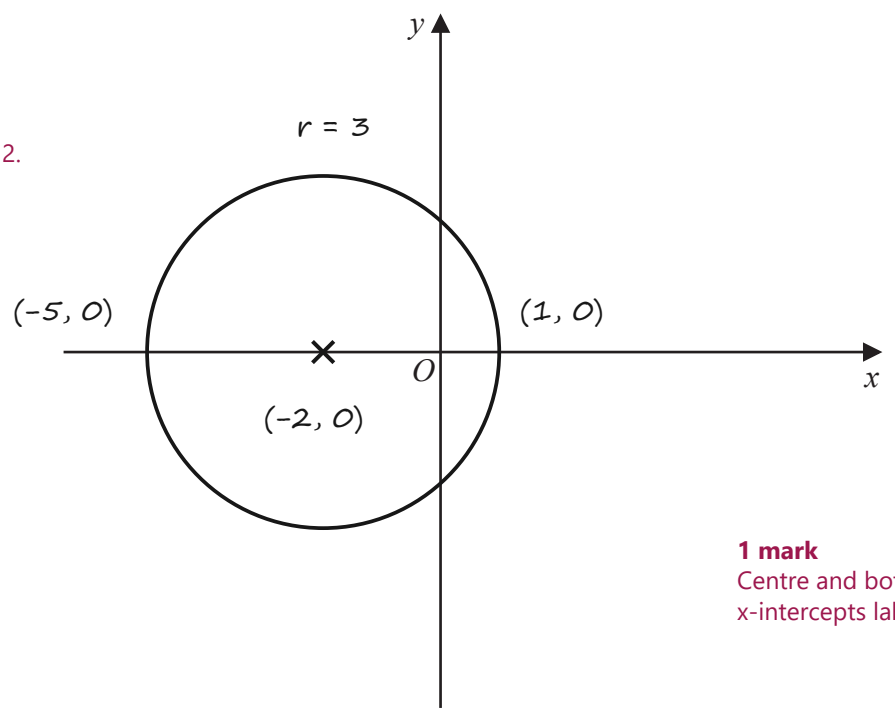
(a) On the grid above, sketch the graph of $y = -f(x)$

(1)

C is the circle with equation $x^2 + y^2 = 9$

The circle **C** is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ to give circle **S**.

(b) Draw a sketch of circle **S**.
Label with coordinates the centre of circle **S** and any points of intersection with the x-axis.



1 mark
Centre moves left 2.

1 mark
Radius stays 3.

1 mark
Centre and both
x-intercepts labelled.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

T is a circle with centre (0, 0)
P is the point on **T** with coordinates (12, −9)

(c) Find an equation of the tangent to **T** at the point *P*.

Radius gradient = $\frac{-9}{12} = \frac{-3}{4}$

1 mark
Gradient of OP.

Tangent is perpendicular: gradient = $\frac{4}{3}$

1 mark
Negative reciprocal.

Through P(12, −9): $y + 9 = \frac{4}{3}(x - 12)$

$y = \frac{4}{3}x - 25$

1 mark
Correct line through P.

(3)

(Total for Question 21 is 7 marks)